How Voice Call Technology Posts Security in Threats in 4G LTE Networks

Guan-Hua Tu[1], Chi-Yu Li[1],
Chunyi Peng[2], Songwu Lu[1]

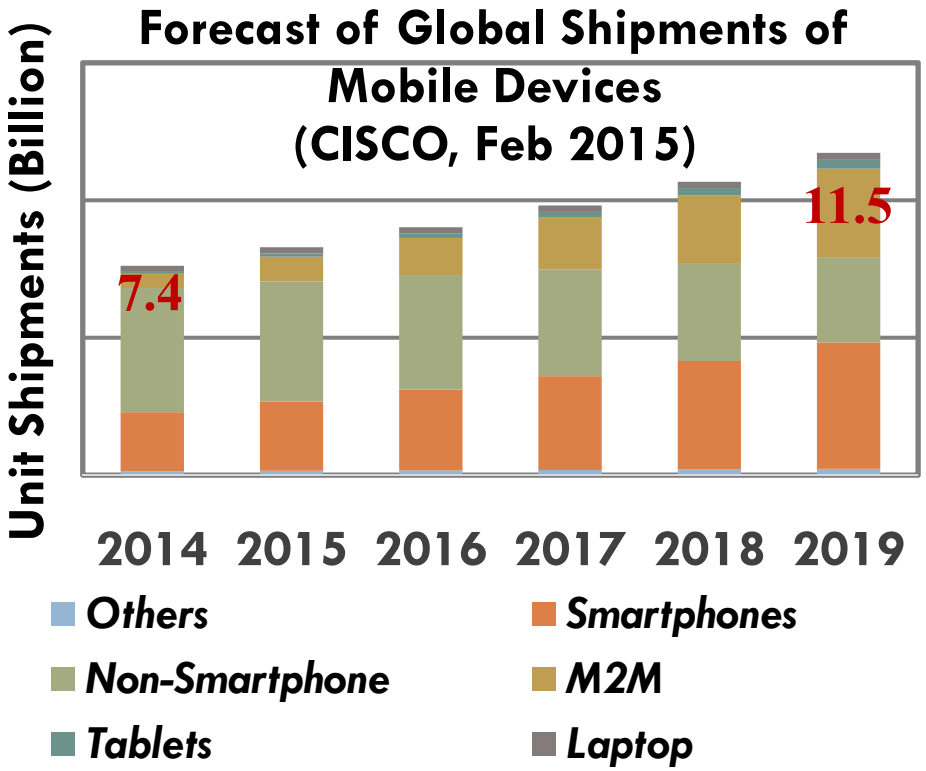


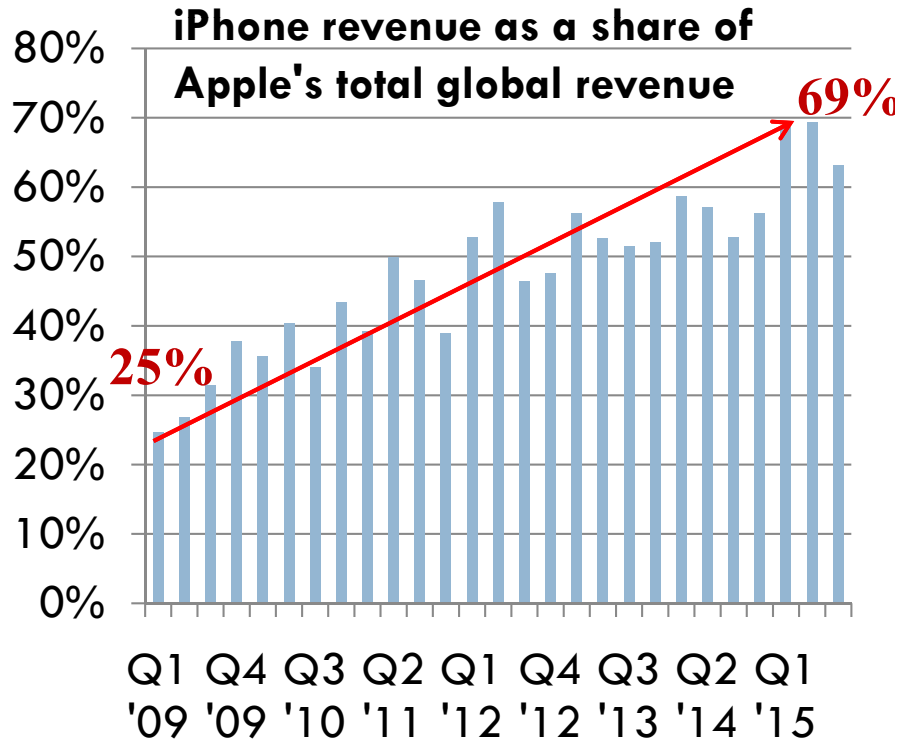
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2: The Ohio State University



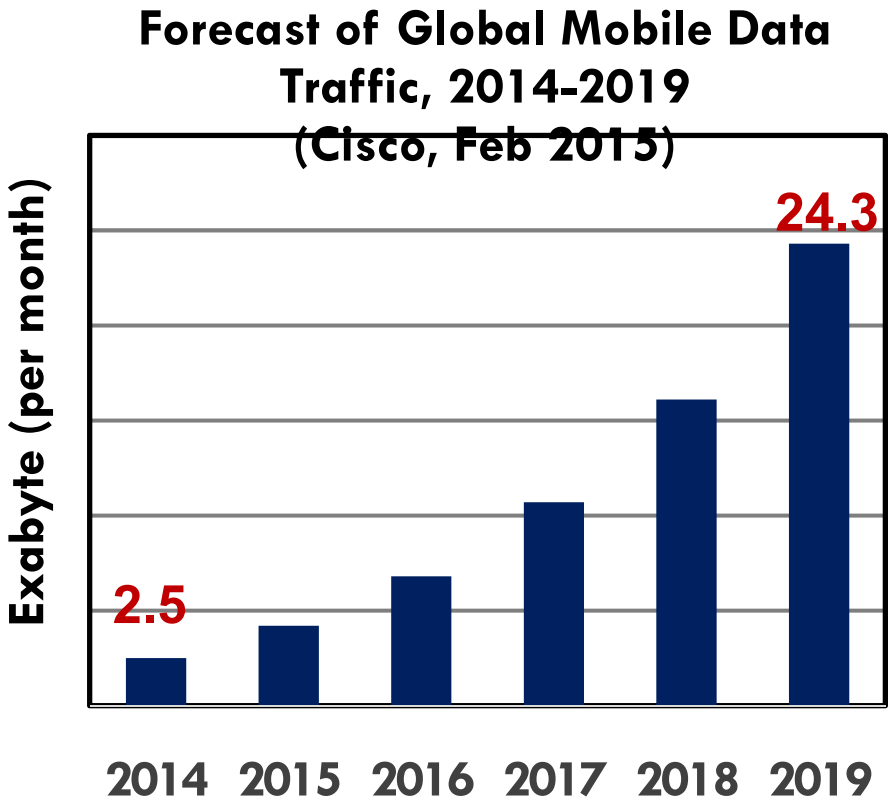
Ubiquitous Mobile Network Services

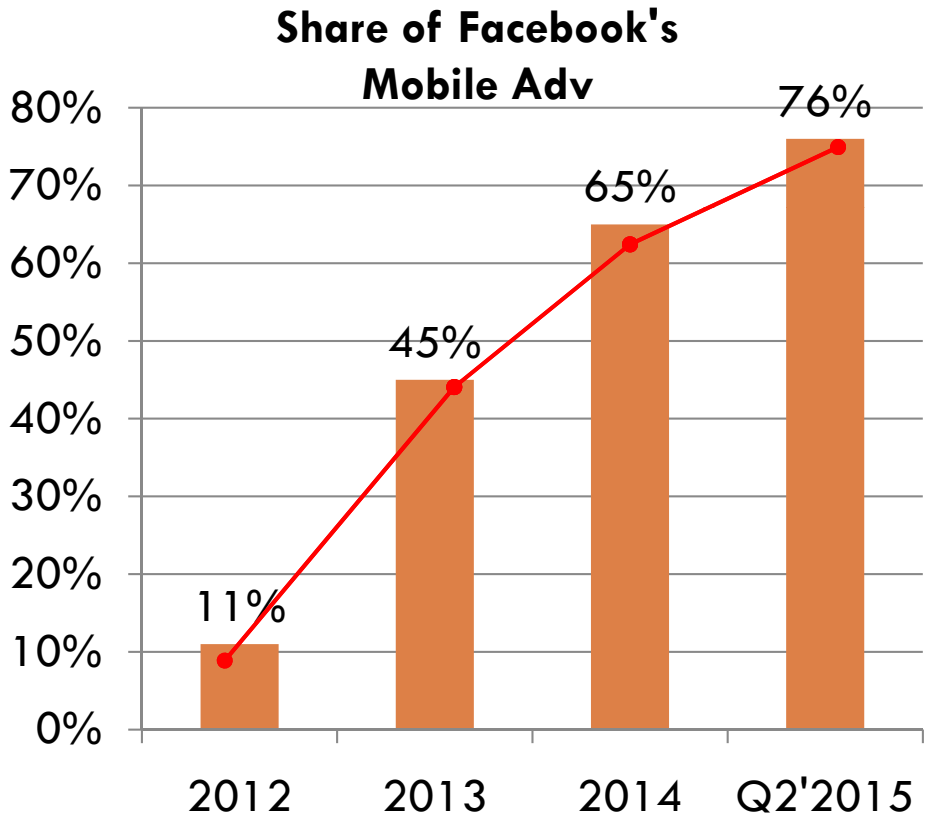
- Global voice and data services
- More than 7 billion mobile-connected devices





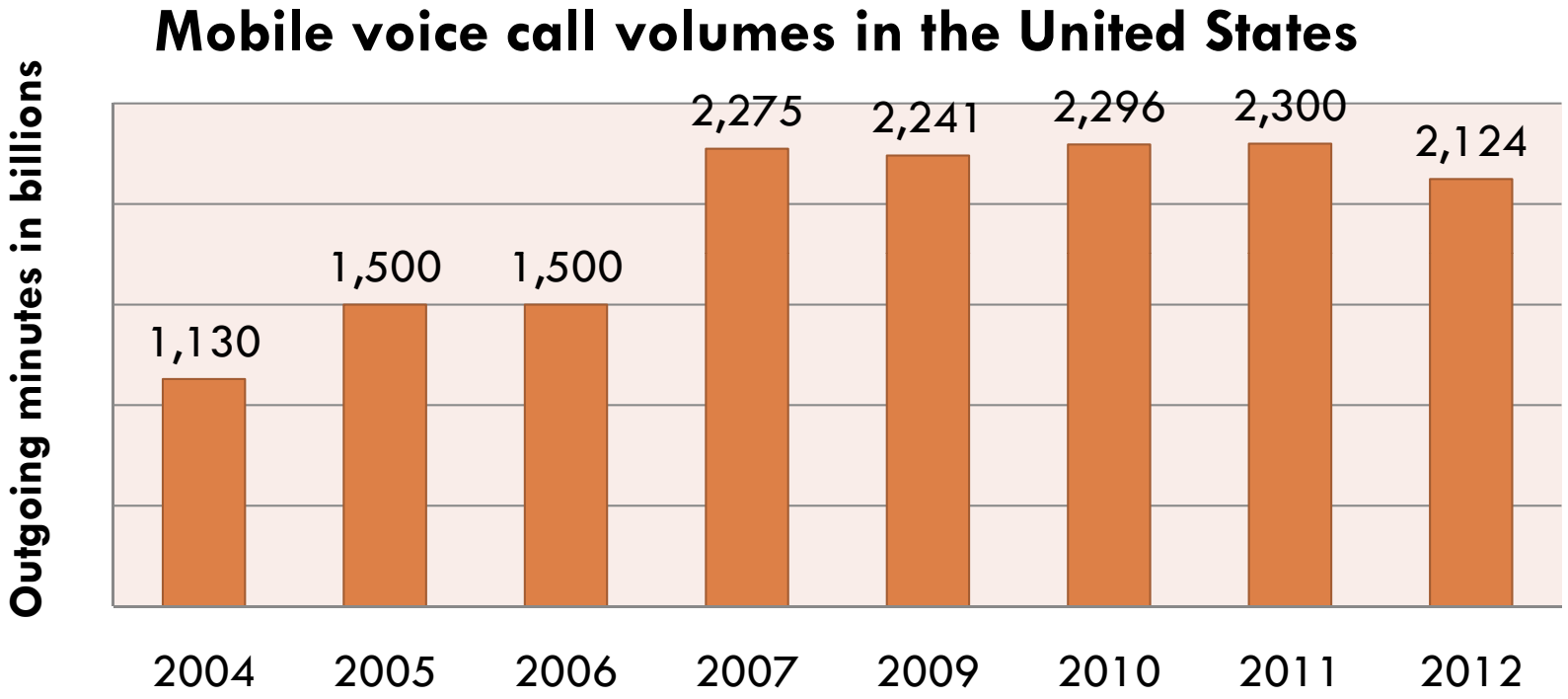
Increasing Popularity of Mobile Data Service





3 times faster than global fixed IP traffic from 2014 to 2019 !!

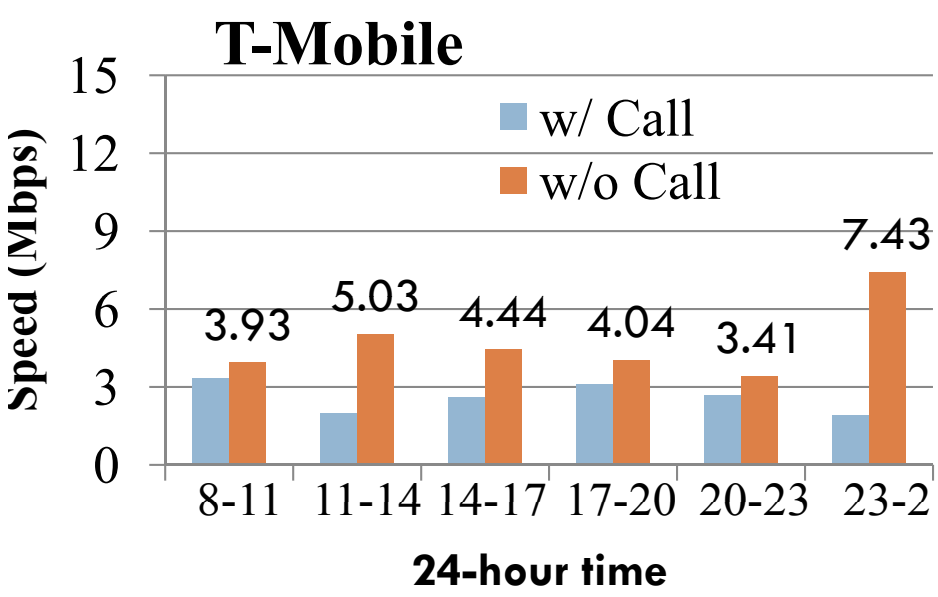
Voice or Data Service? Which one is more important to carriers?

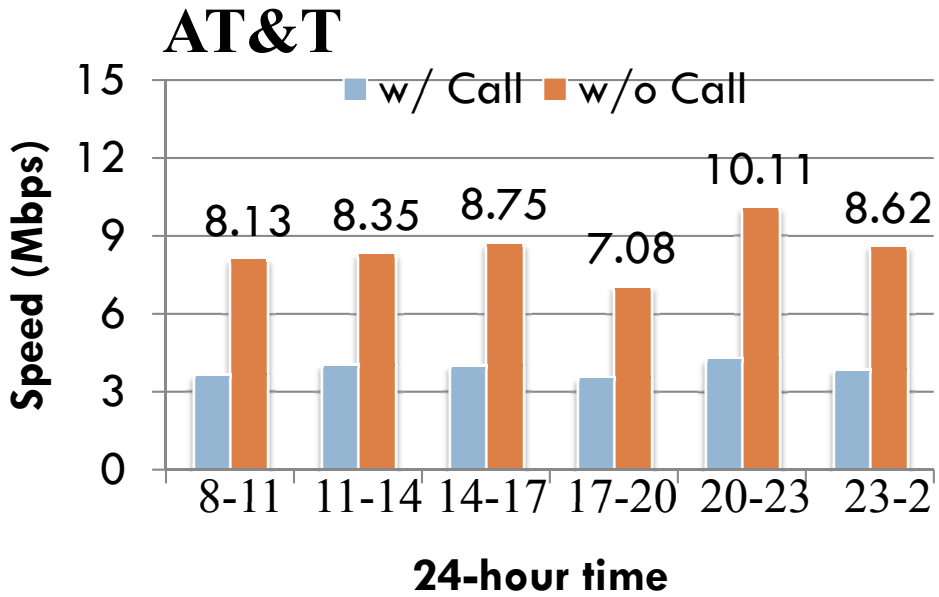


- Data-Service-First?

- **Unfortunately, it is never the case for last three decades.**
- **To provide user with ``carrier-grade voice``, there is nothing non-disposable or non-sacrificed !!**

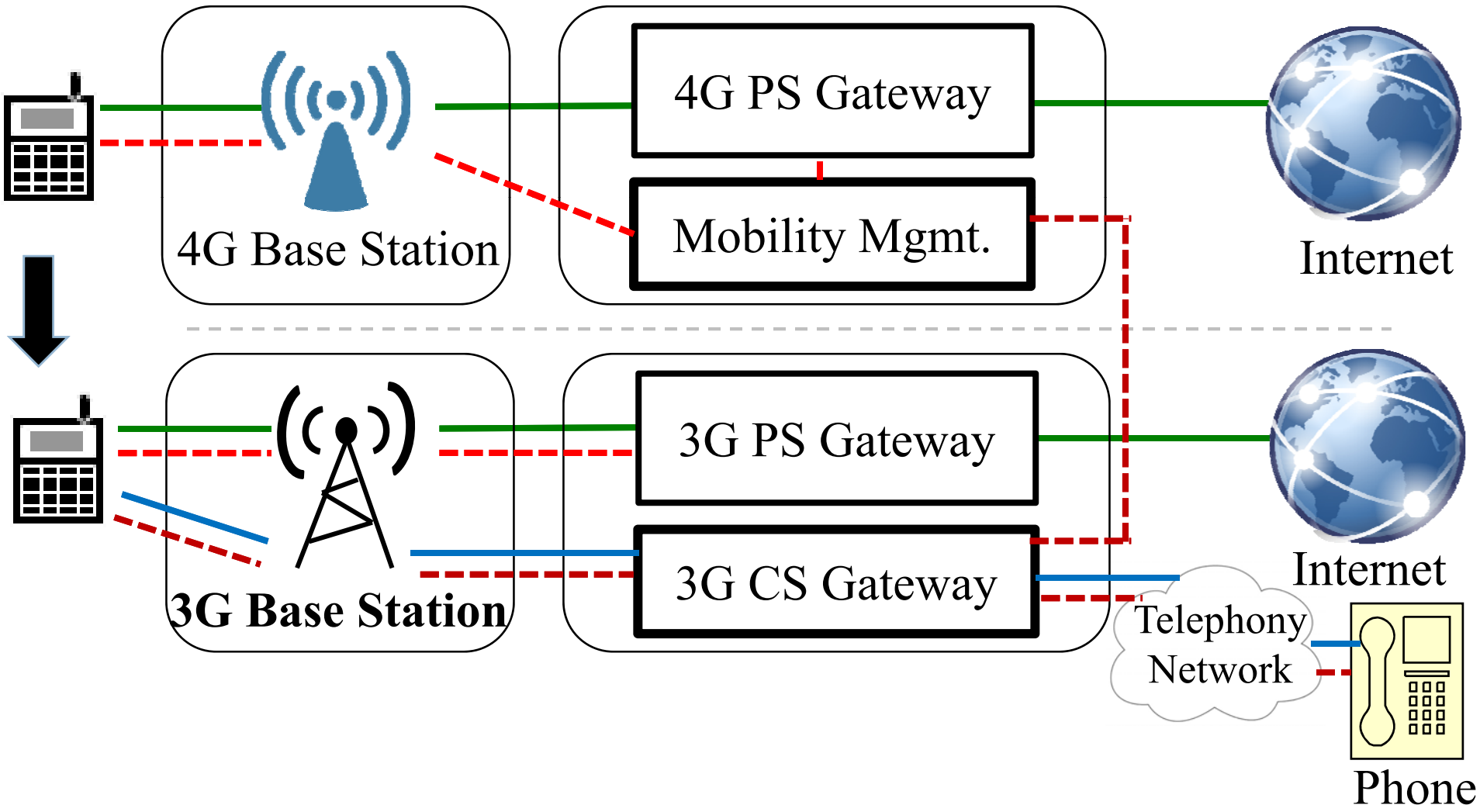
74.8% Performance Downgrade During Call in 3G [Sigcomm'14]



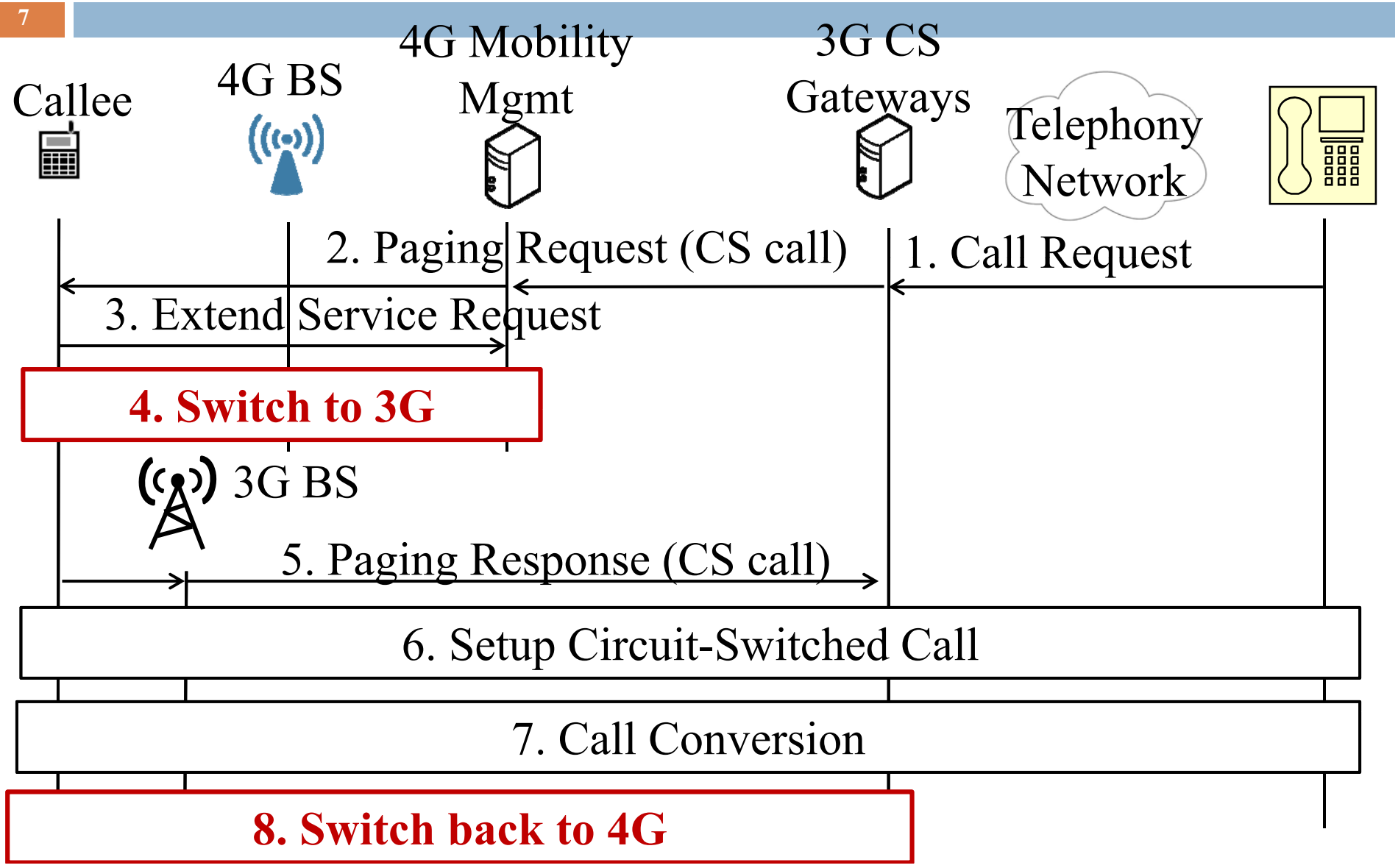


- How about 4G voice service?

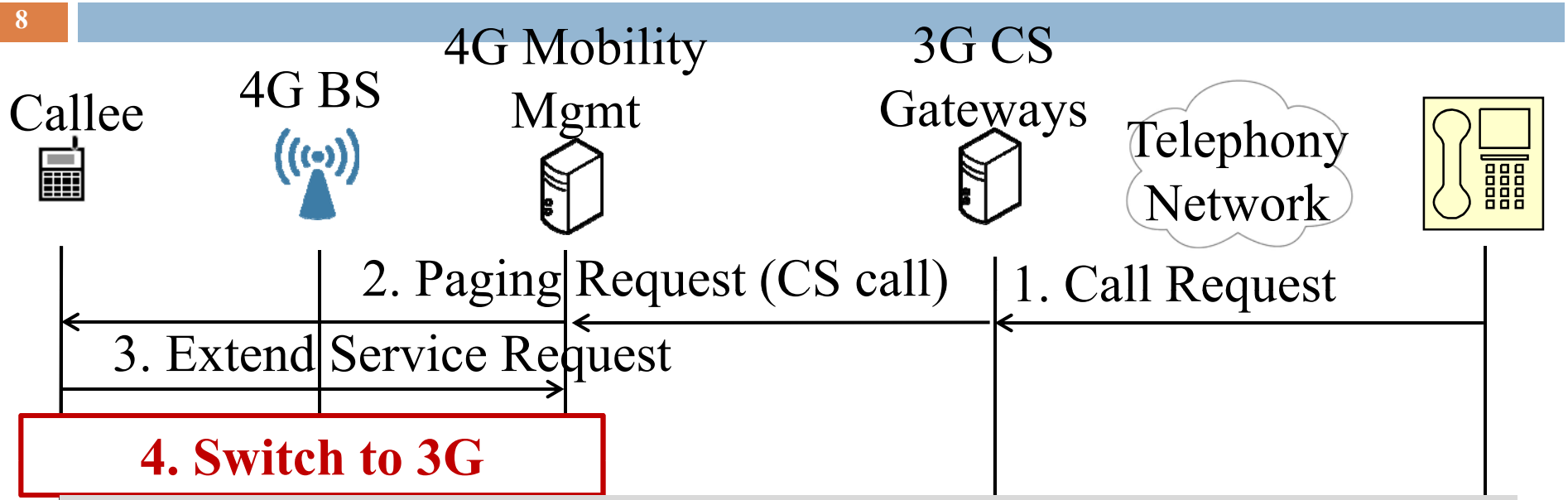
CSFB (Circuit-Switched Fallback)



CSFB: Incoming Call Flow

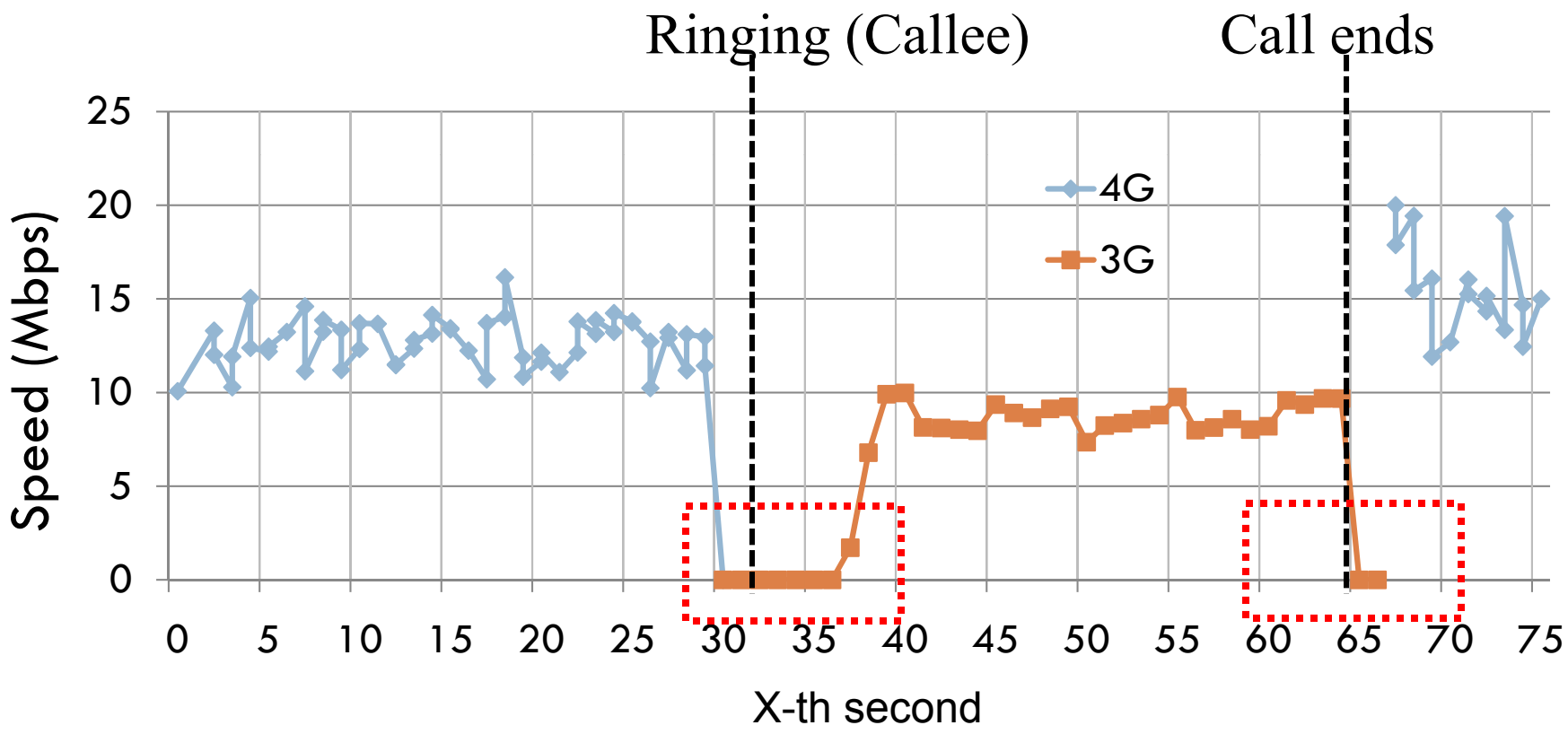


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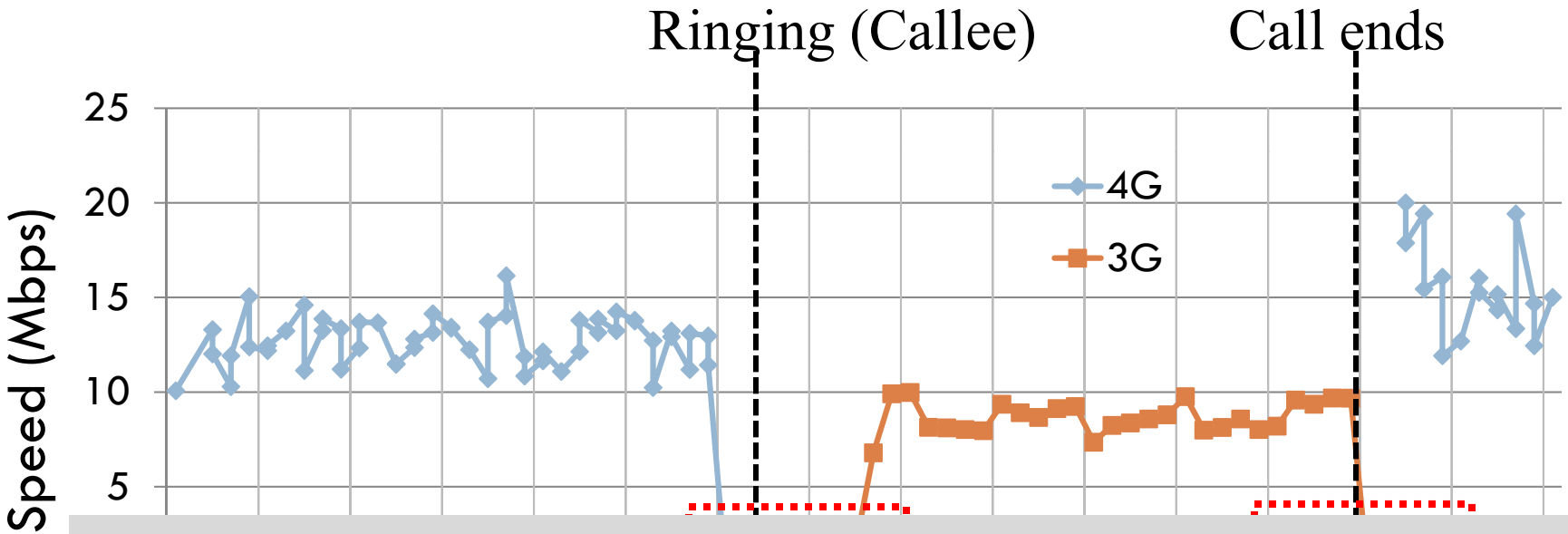


- It seems good to carriers and mobile users
  - Carrier reserves 4G LTE network for high-speed demand users
  - Carrier fully utilizes the existing and well tested 3G infrastructure
  - Users are ONLY switched to 3G when needed
- What is wrong??

An Illustrative Example 1

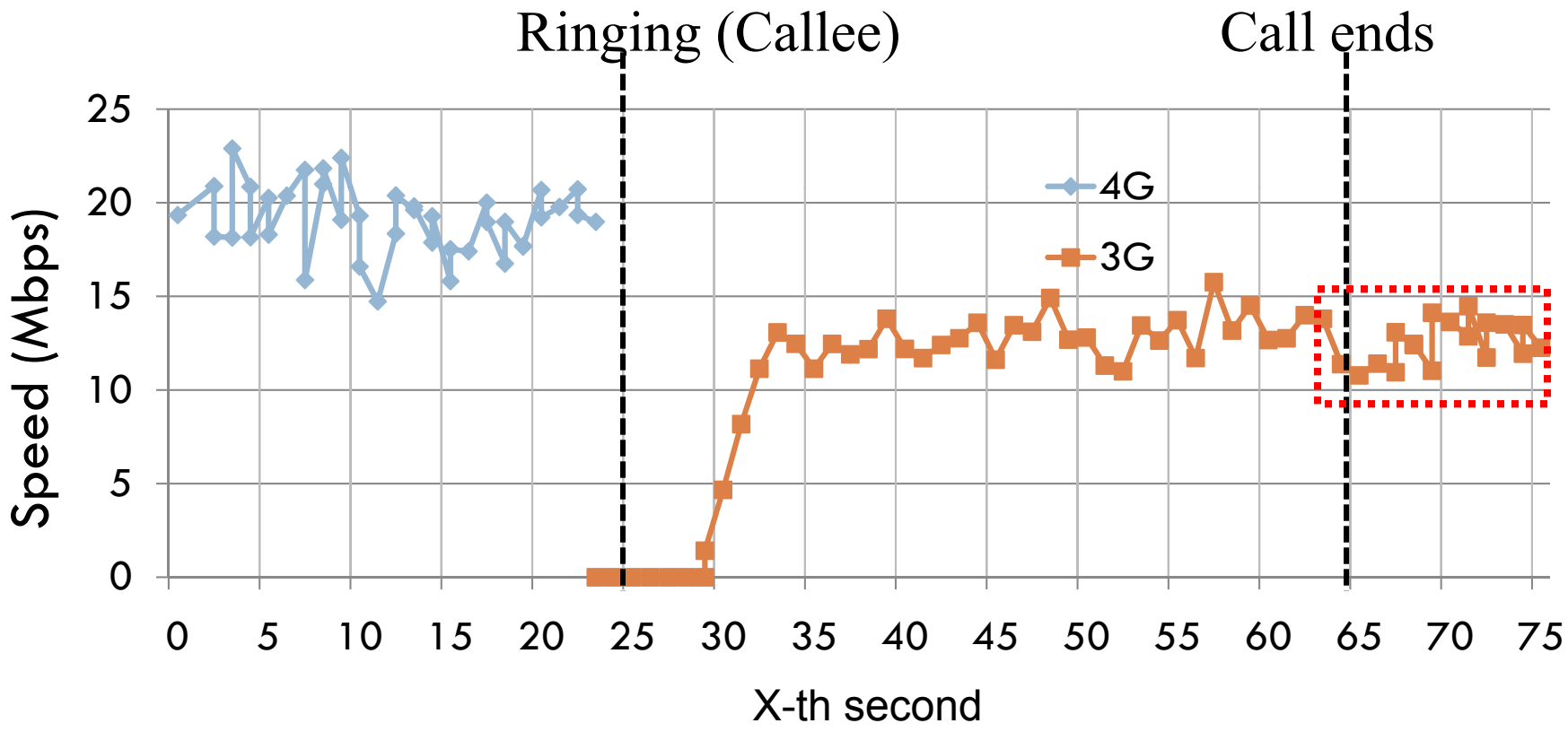


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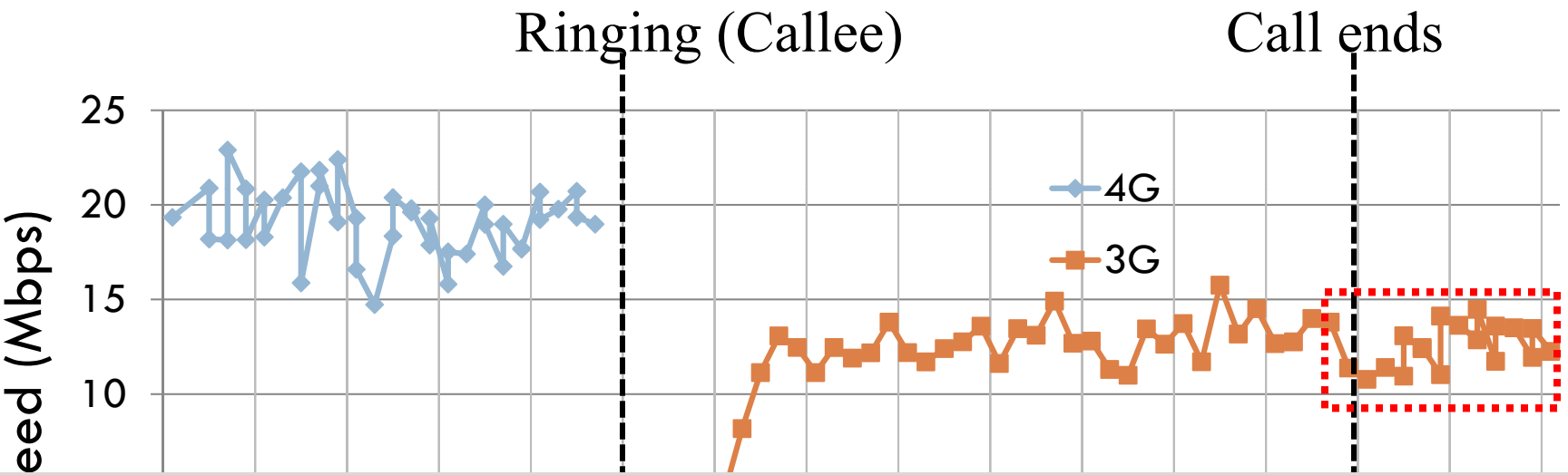


- **Two observations for one tier-one US carrier, OP-1 (35% market share)**
  1. **While 4G<->3G occurs, data service is interrupted.**
  2. **Carrier proceeds CS call service prior to resuming the data service (long data suspension time, 6-7s)**

An Illustrative Example 2



An Illustrative Example 2



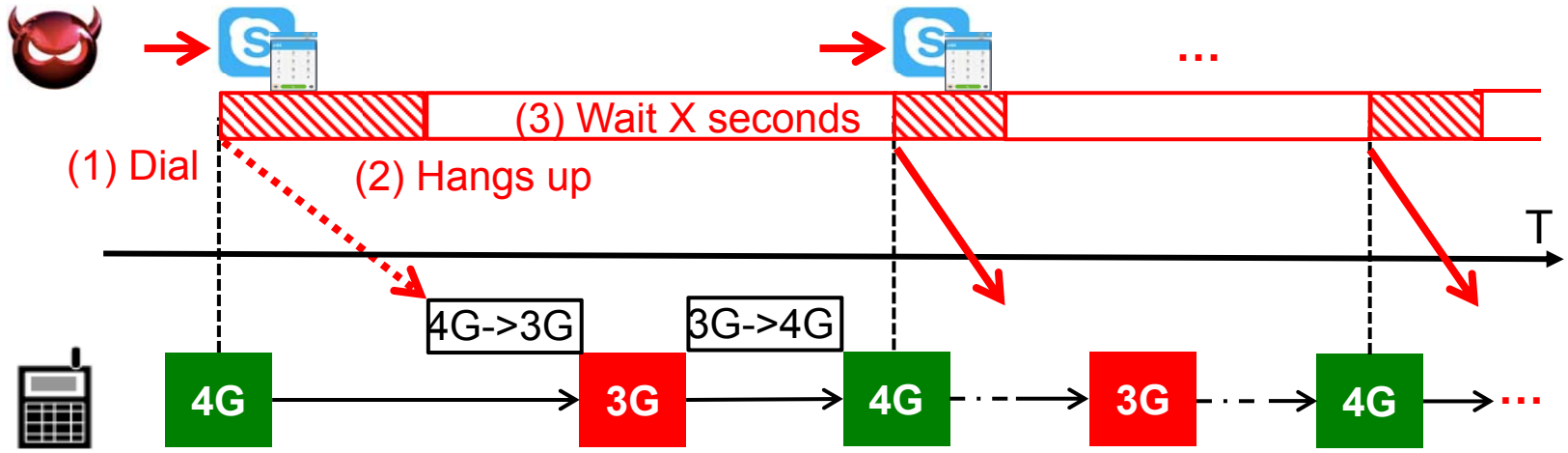
- **Observations for one tier-one US carrier, OP-2 (10% market share).**
  - **Similar to previous one => CS voice first, long data disruption**
  - **After the call ends, the user will NOT be switched back to 4G unless there is no PS traffic.**
  - **Even there is no PS traffic, carrier takes several to tens seconds to switch users back to 4G**

Security Vulnerabilities

- V1: A 4G→3G switch can be triggered by the caller without the callee's consent
- V2: Carrier proceeds CS voice call setup priori to PS data service recovery
  - Long data service disruption
- V3: The subsequent 3G→4G switch may be deferred or disabled by malicious callers
  - E.g., calling the victim again before carrier moves he/she back to 4G

Ping-Pong Attack (cont.)

- What if we conduct the successive dialing-hang-up calls towards victims?

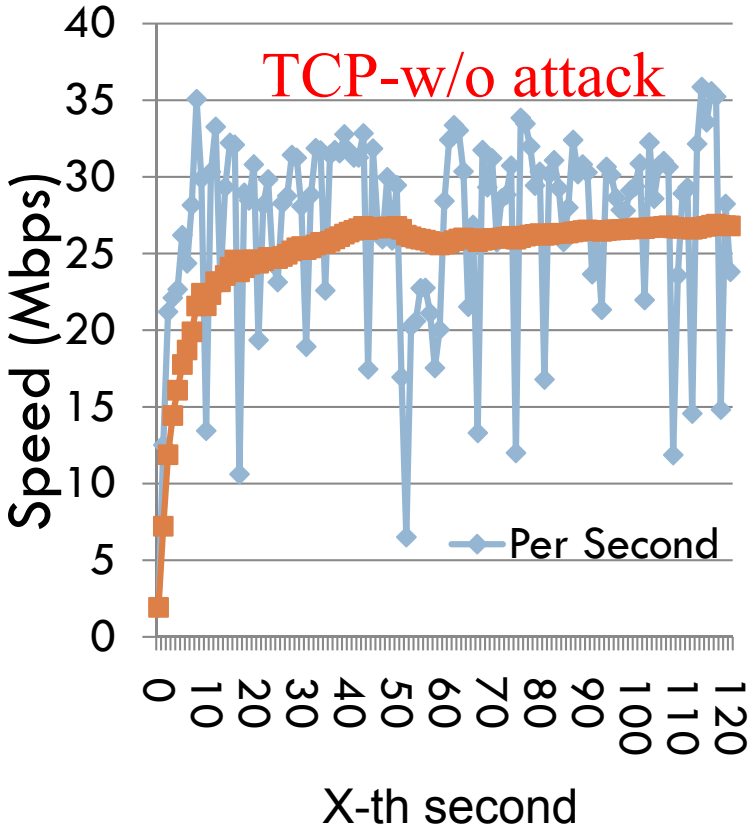


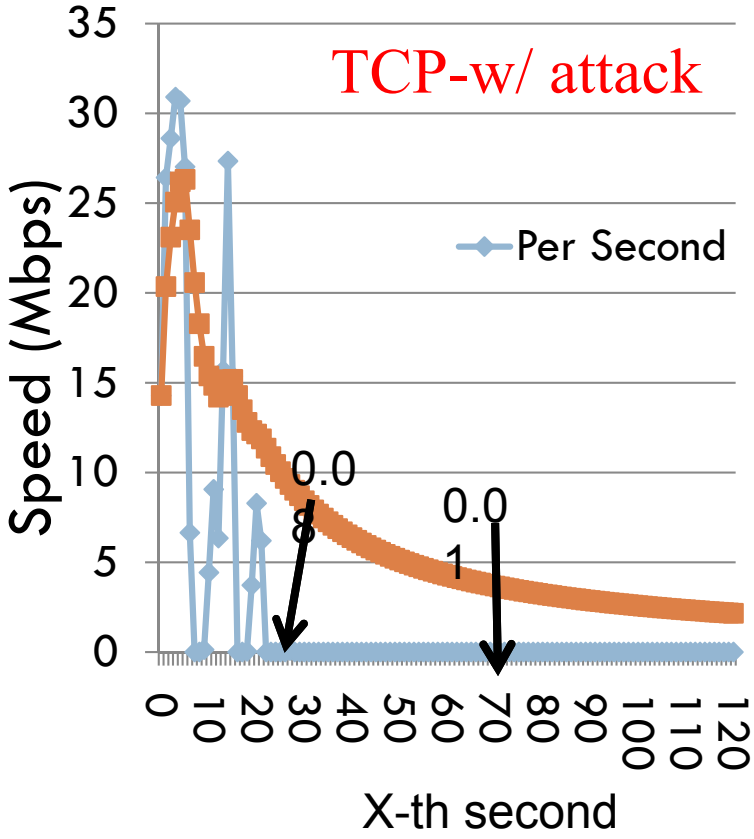
- **Step 1: Attacker calls CSFB victim to trigger 4G->3G switch**
- **Step 2. Attacker cancels the call before CS call is ready for ringing**
- **Step 3. Attacker waits for X seconds** (allows victims to be switched back to 4G) **and repeats Step 1**

The detailed configurations of hanging the outgoing call up and X seconds to reach longest data disruption time varied with carriers. Please to refer to our paper

Ping-Pong Attack (cont.)

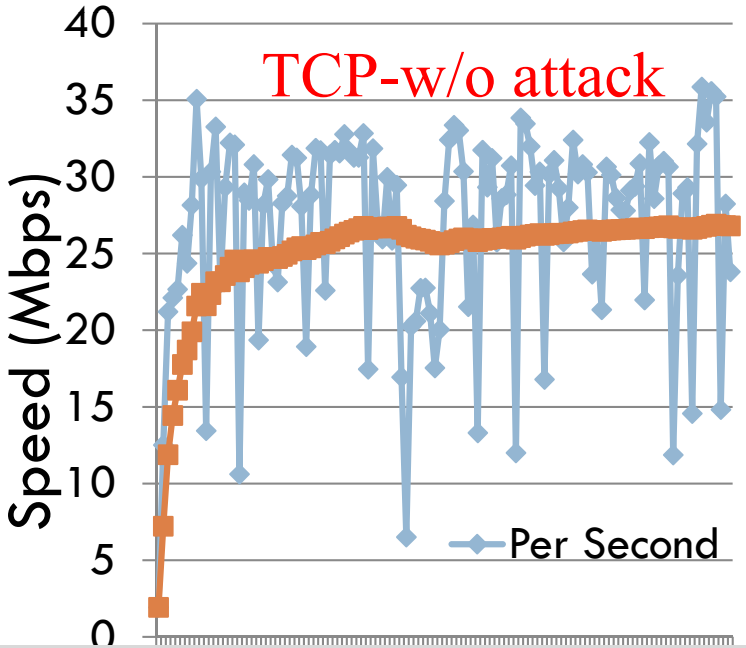
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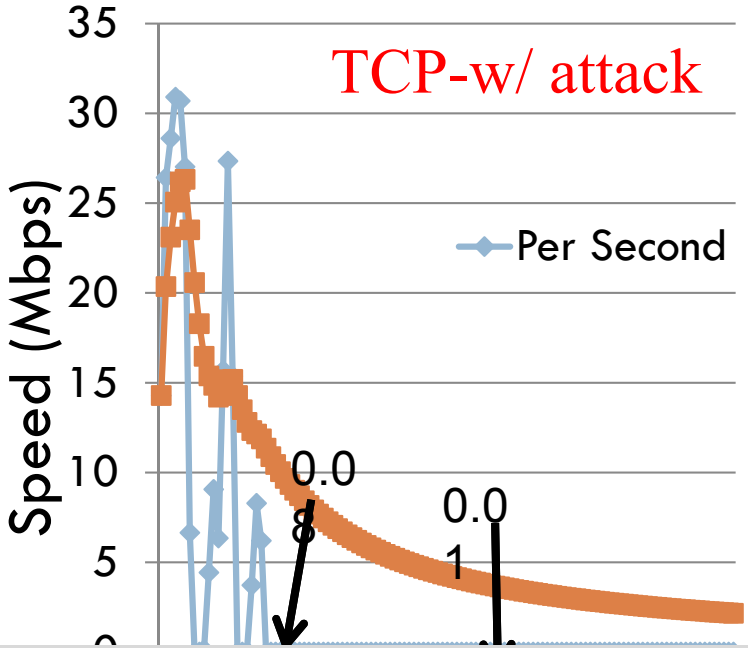




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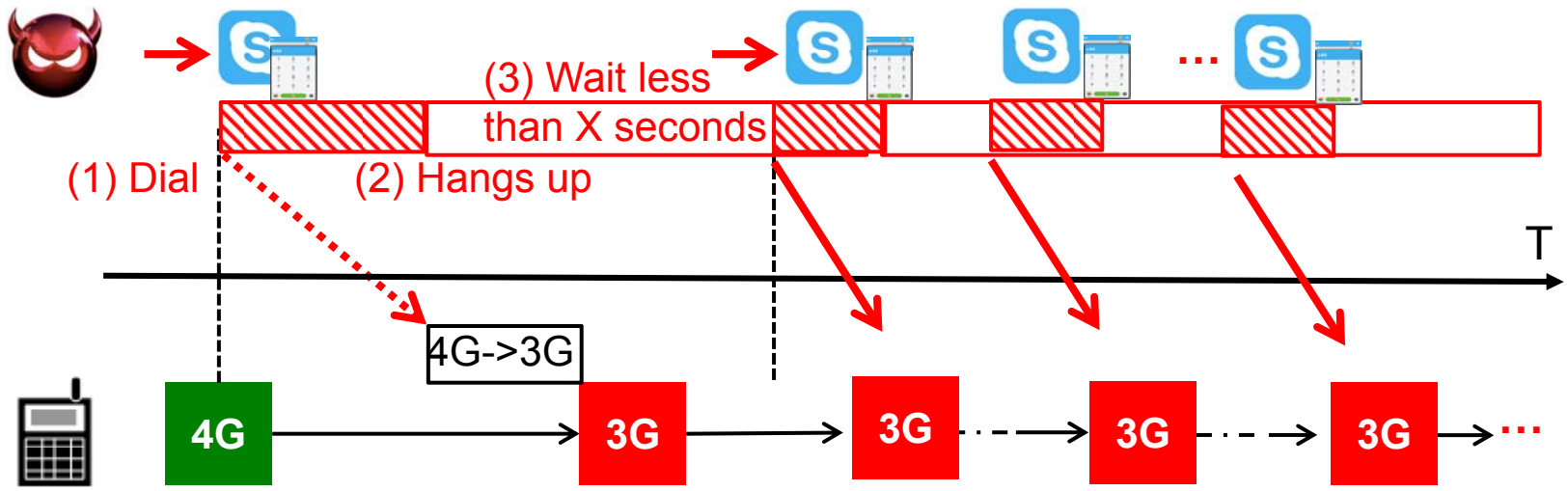




- **Affect TCP-based applications.**
  - **Reduce speed from 31Mbps to 0.08 Mbps in 30s.**

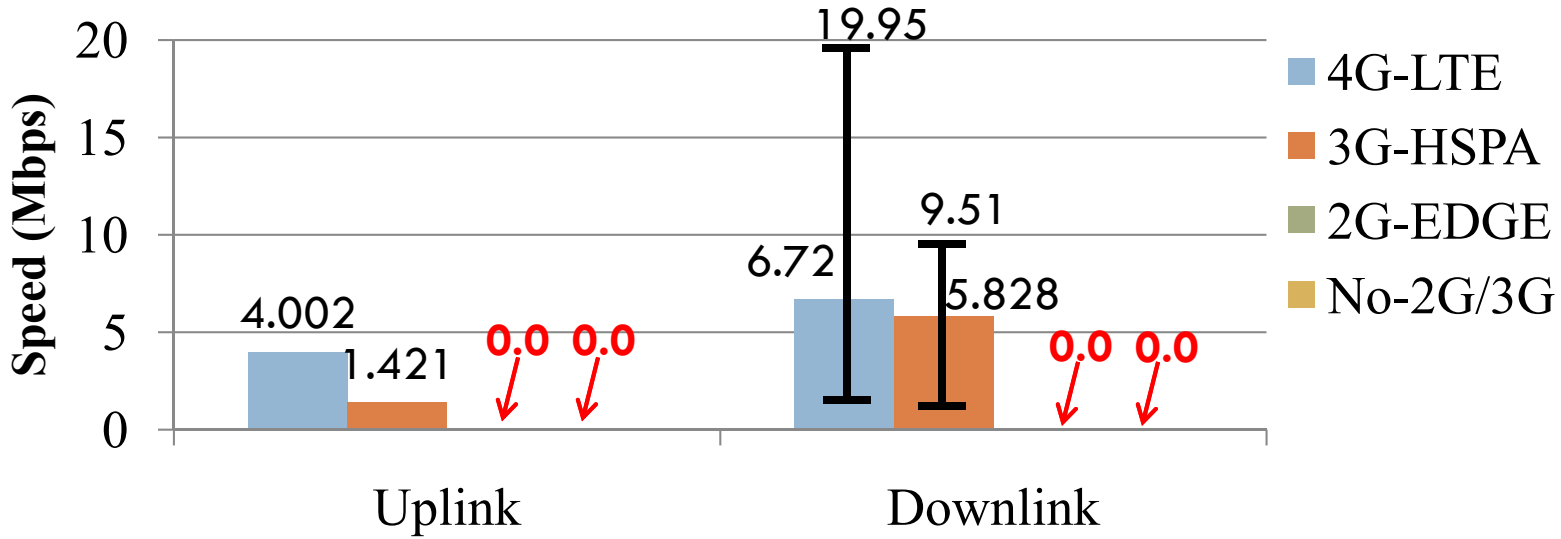
Ping-Pong Attack (cont.)

- What if attacker redials before victim is switched back to 4G?



Ping-Pong Attack (cont.)

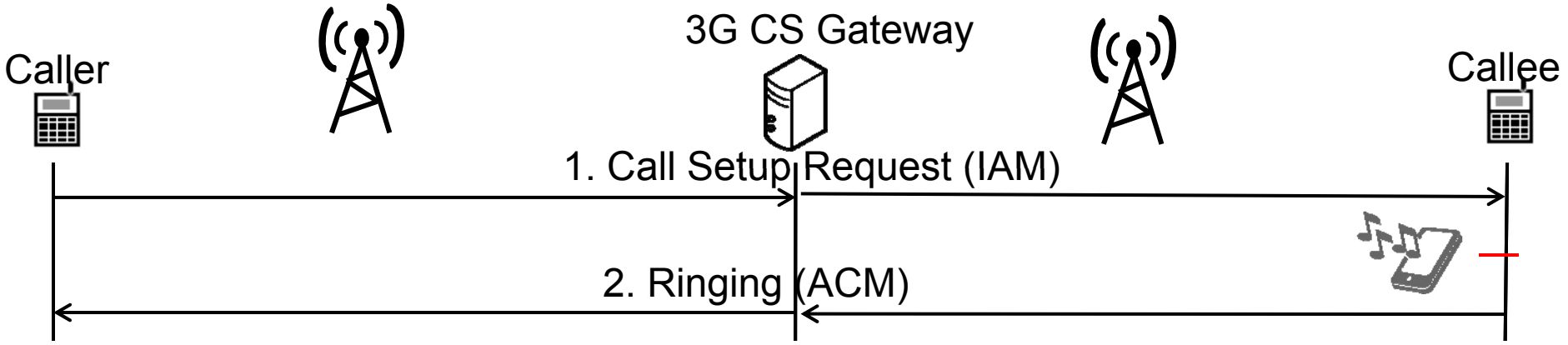
- Stuck in 3G if victim is lucky



- **When victim gets stuck in 3G, the performance is downgraded up to 76% (uplink) and 53% (uplink)**
- **When victim gets stuck in 2G, data service is NOT available (non-current voice and data service in 2G, e.g., AT&T and T-Mobile)**

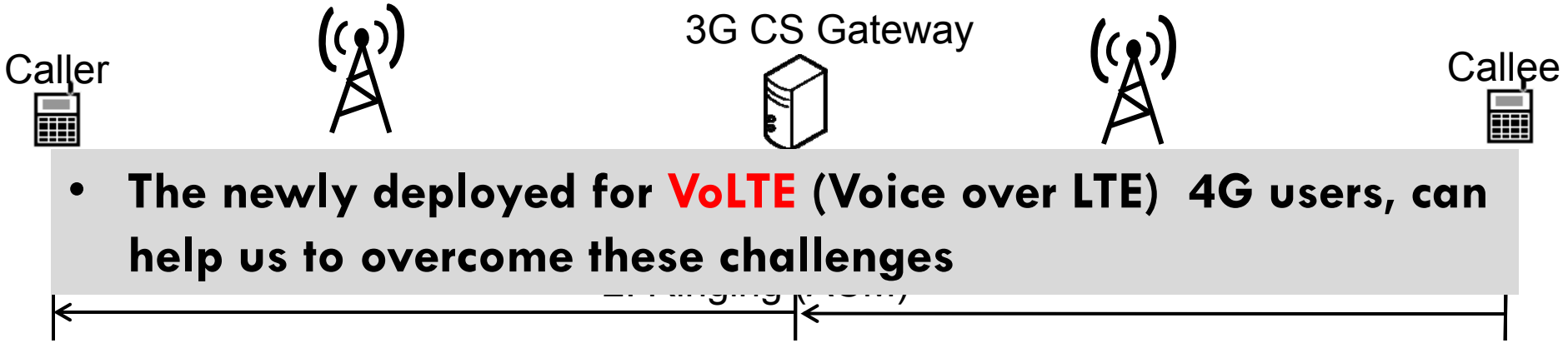
Challenges

- How to successively switch the victims to 3G and victims are unware (e.g., ringing) of the malicious incoming calls without fixed timer approaches (dialing, hang up timer)?
- Issues:
  - Carrier will NOT notify the caller that the callee has been switched to 3G networks.
  - Once callee is switched to 3G, the call setup procedure is immediately proceeded and callee's phone may ring at anytime



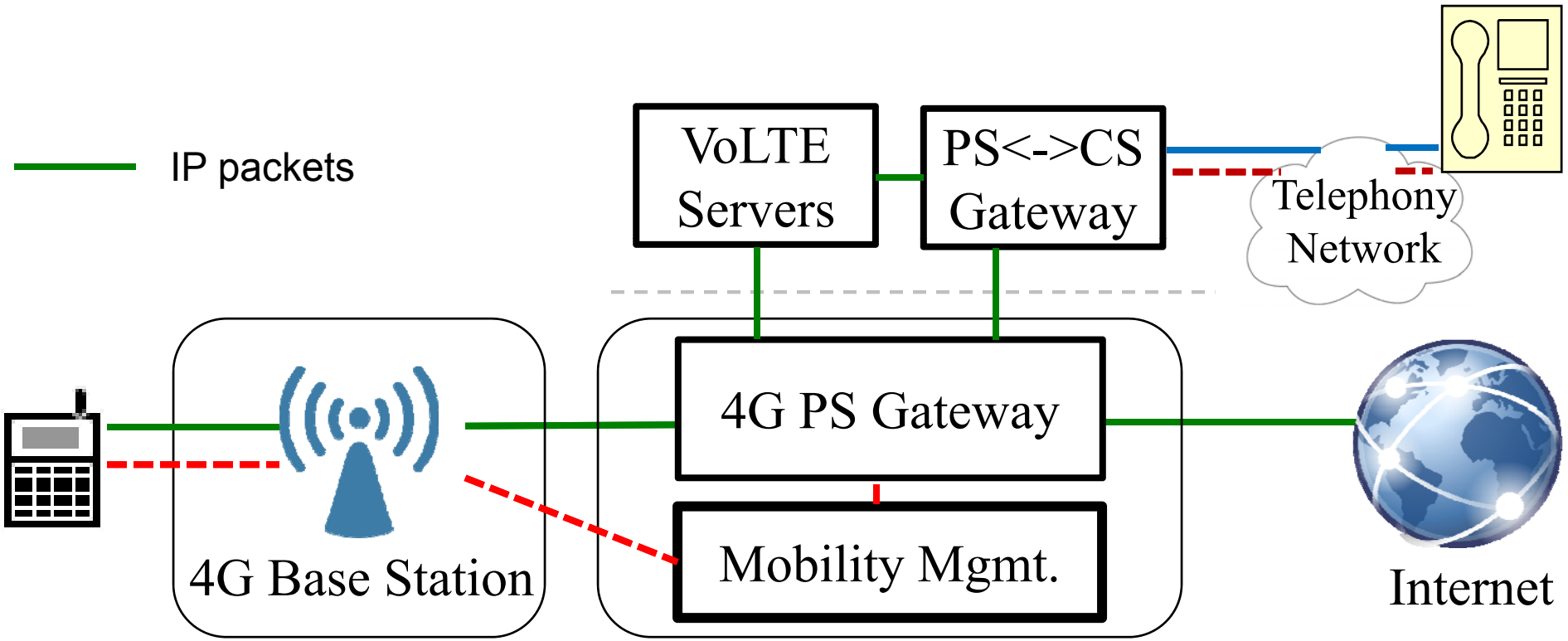
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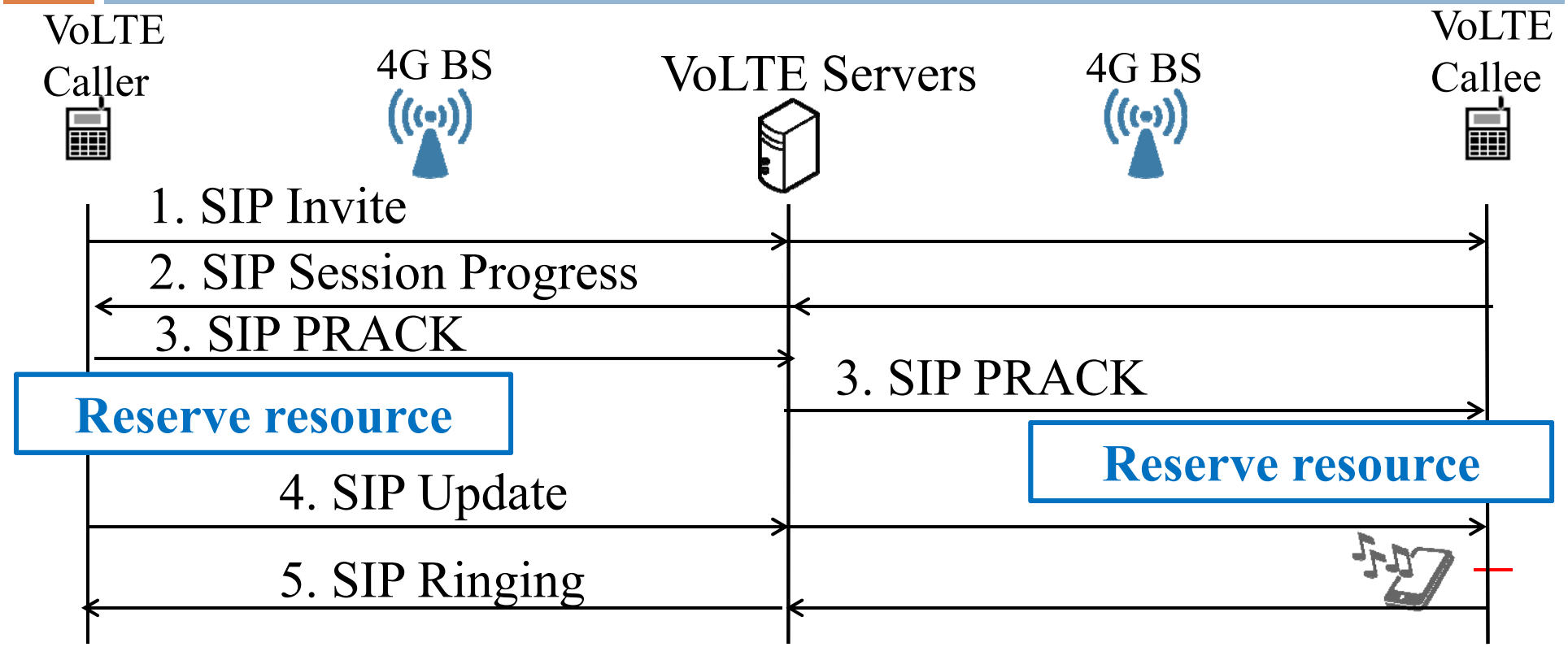
- **The newly deployed for VoLTE (Voice over LTE) 4G users, can help us to overcome these challenges**

VoLTE (Voice over LTE)



- **A VoLTE is a SIP (Session Initiation Protocol) call which is revised for the use scenario in the cellular networks**
- **What does it mean??**

VoLTE Outgoing Call Flow

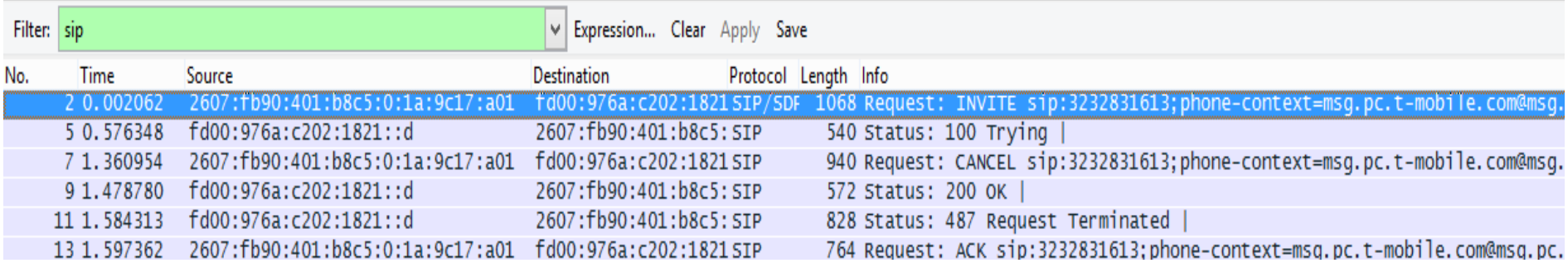


To provide better use experience, the callee's phone is only ringing after the *resources reservation* at both caller and callee finish.

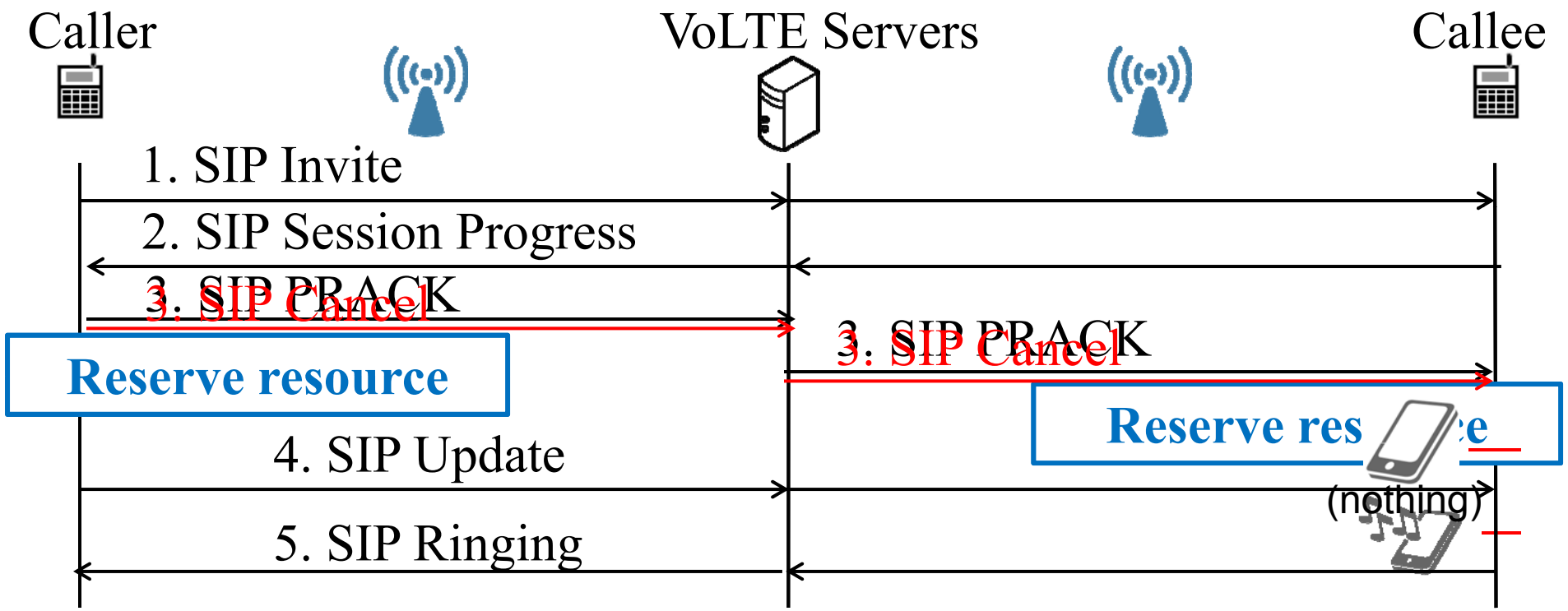
Security Vulnerability

- To support "Carrier-Grade-Voice", the VoLTE adds cellular-specific resource reservation procedure
  - It offers a chance to caller to well control whether the callee's phone can play the ringtone
- Software-based VoLTE client deteriorate this issue
  - Major tier-one US carriers, AT&T, T-Mobile, Verizon deploy VoLTE client as app (e.g., Android app)

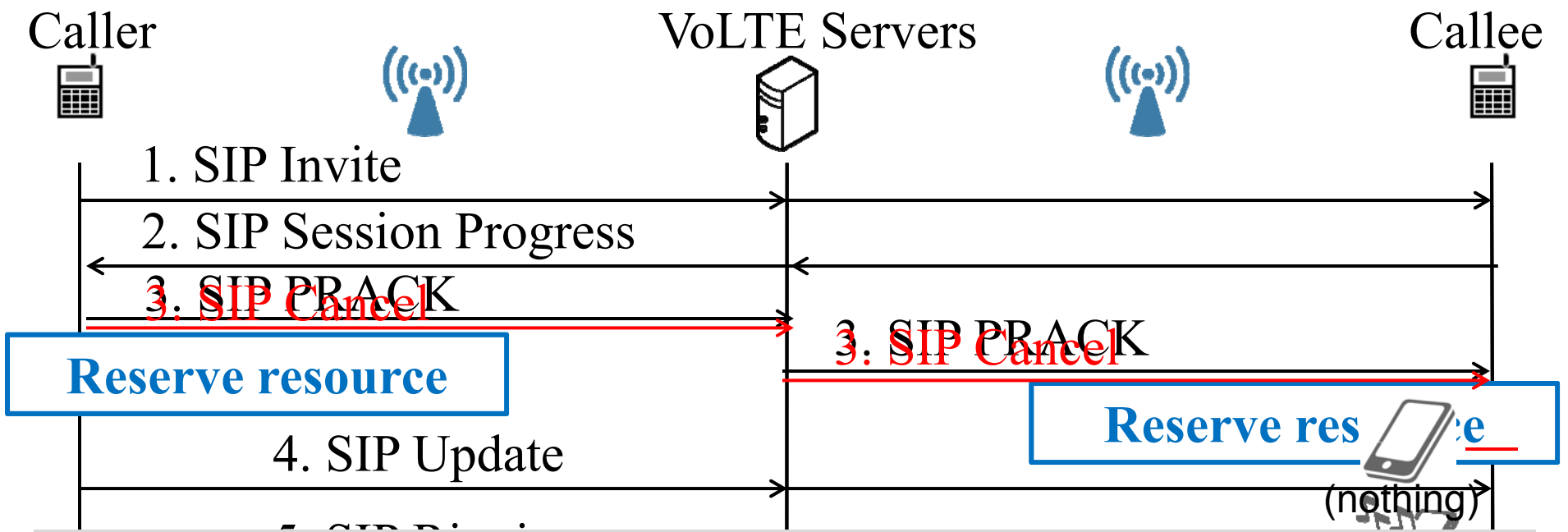
We can monitor all VoLTE signals exchanged between VoLTE client and server through native C APIs on phone, then play tricks.



Malicious Use Scenario



Malicious Use Scenario

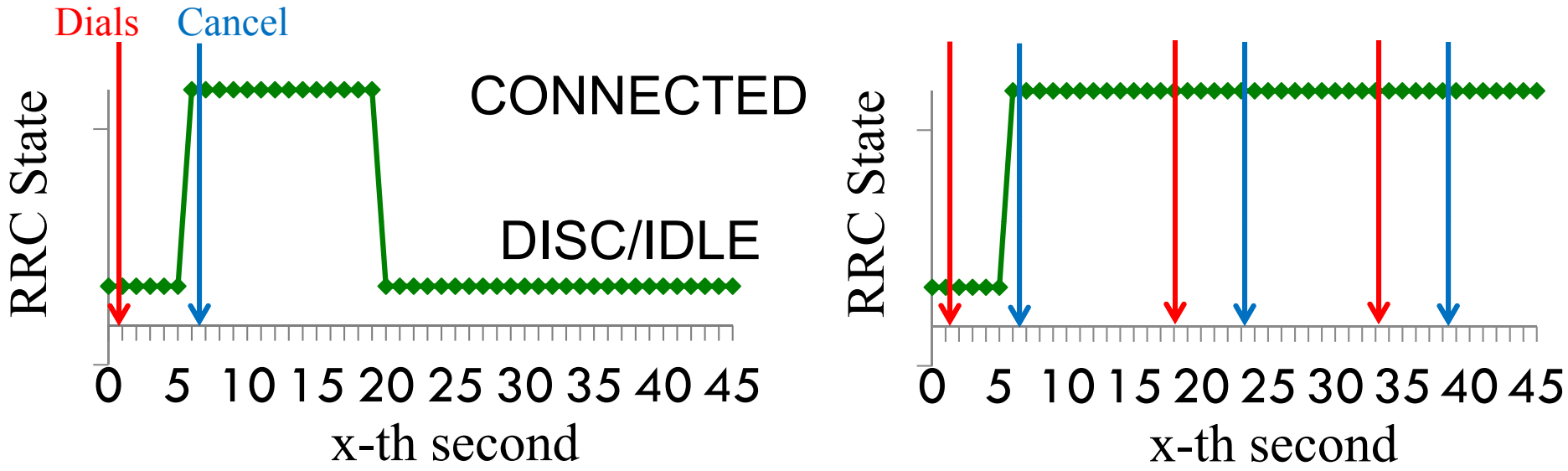


Callee does not get any indication about this *incomplete* call from phone:

(1) No ringtone, vibration, call history or calling number

(2) Android only delivers incoming call event to app after the call is ready for ringing.

For VoLTE victims: Induce 8x Power Consumption

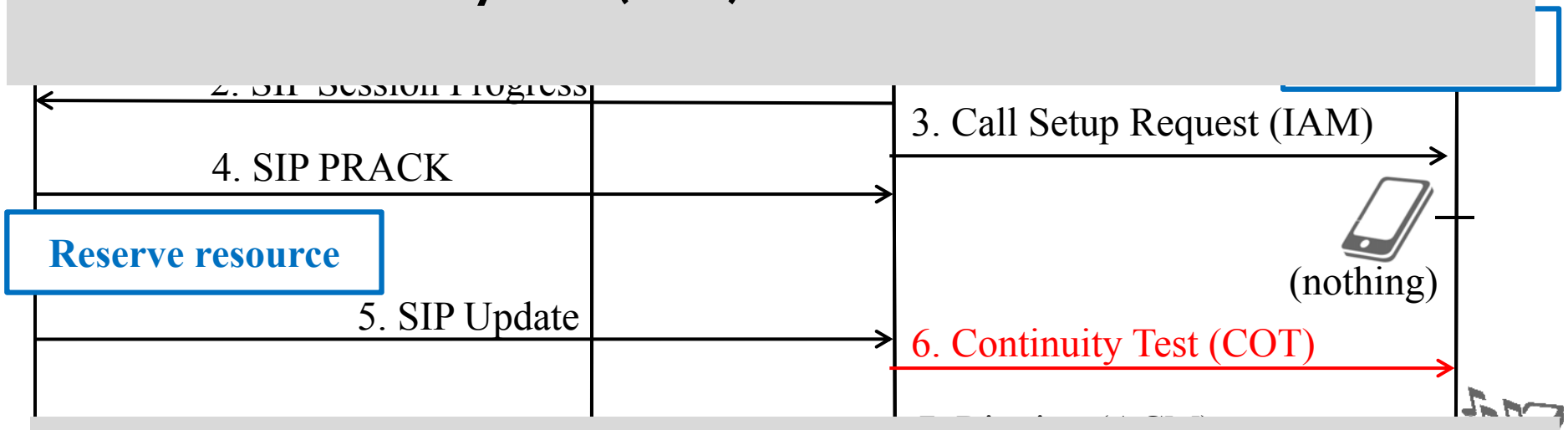


	Attack	No-Attack	Attack/No-attack
Battery usage	**24%**	**3%**	**800%**

(Samsung Galaxy S5)

Apply VoLTE to Ping-Pong Attack

- **To reduce the call setup time, PS<->CS Gateway sends IAM to the CSFB callee before Caller's resource reservation finishes.**
- **To suppress the ringtone at callee before caller is ready, VoLTE enables Continuity Test (COT) function in IAM**



- **The design allows attackers to well prevent the CSFB victim from knowing ping-pong attack.**
  - **Simply cancels call before SIP Update is sent**

Bypass Current Defenses

- **Phone-based call-blocking or spam detection applications**
  - Not working; These apps only receive the notification from mobile OS (Android) after the incoming call is ready for ringing
- **Carrier-based call blocking service**
  - Not working; it requires users to provide the phone numbers being blocked
  - However, victims do not have such information

Recommended Solutions

- Unawareness Prevention
  - Phones should not automatically agree with the 4G->3G switch request from carrier without users' consents

In all 4G CSFB phone implementations, they always immediately reply "Extended Service Request"with "YES to Switch to 3G" after paging request is received.

  - Hardware (modem) should provide software (Mobile OS) with all information of incoming calls no matter it is a complete call or not

Summary

- CSFB lies in seemingly sound design decisions from the functional correctness and performance standpoints of voice services
  - For shorter call setup time, switch callee to 3G without their consents
  - Setup 3G CS call before recovering data service
  - ….
- Such choices may lead to more severe attacks against data services
- Voice-service-first design princple requires revisiting.

Questions?